

FACE RECOGNITION ATTENDANCE

Data Cleaning & Visualization Project Report

Prepared as an internship data analysis submission

1. Project Overview

This project demonstrates a complete data-cleaning and visualization workflow for attendance records associated with a face-recognition attendance system. The workflow uses Python, Pandas and Matplotlib to transform raw attendance records into a clean dataset, calculate attendance metrics and communicate findings through charts and a dashboard.

2. Objectives

- Identify and handle missing values.
- Detect and remove duplicate attendance records.
- Standardize student names, departments, attendance status, dates and times.
- Calculate student-wise and daily attendance percentages.
- Identify students below the 75% attendance threshold.
- Create visualizations and a summary dashboard.

3. Tools & Technologies

Python, Pandas, NumPy, Matplotlib and Jupyter Notebook.

4. Data Cleaning Process

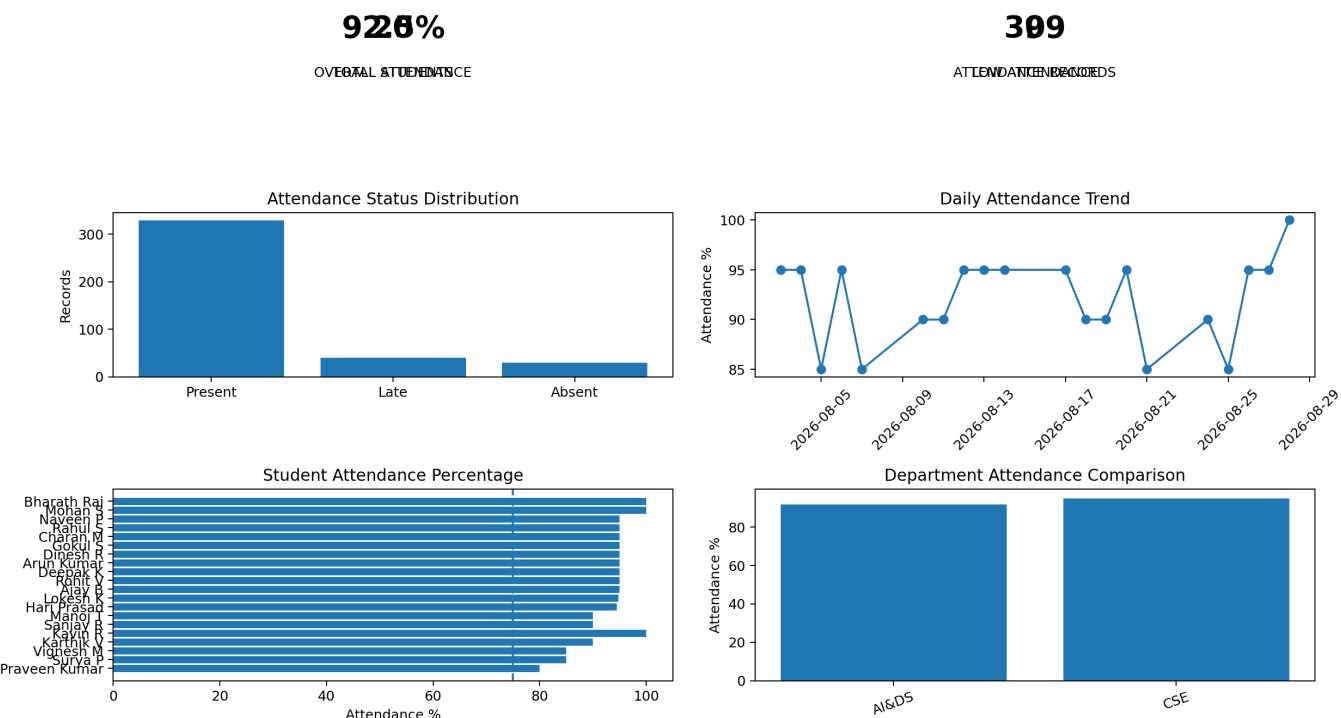
The raw dataset was inspected for missing values and duplicates. Text fields were standardized, dates and times were converted into consistent formats, duplicate student-date attendance records were removed, and invalid status values were excluded. The resulting dataset was saved as attendance_cleaned.csv.

5. Key Results

Metric	Result
Total students	20
Clean attendance records	399
Present records	329
Late records	40
Absent records	30
Overall attendance rate	92.48%
Students below 75%	0

6. Dashboard

FACE RECOGNITION ATTENDANCE DATA ANALYSIS DASHBOARD



7. Insights

The analysis shows an overall attendance rate of 92.48%. The highest individual attendance in the sample is Bharath Raj at 100.00%, while the lowest is Praveen Kumar at 80.00%. The department with the highest attendance percentage in the sample is CSE at 94.95%.

8. Conclusion

The project successfully demonstrates how attendance data can be cleaned, analyzed and visualized. The workflow can be connected to a real attendance database or API generated by a face-recognition system. The dashboard provides a simple way to monitor attendance trends and identify students who may require attention.